

PRA_YAS 2.0



Chapter Name – Work Power Energy

Lecture no. - 02

BY : Praveen Sir

A large blue arrow pointing to the right, with the text 'TODAY'S GOAL' inside it.

TODAY'S GOAL

Work-K.E. Theorem

**Conservative and Non conservative
forces**



Nitin yadav Yadav 6 hr ago

sir aapne bola pull easy hota hai but ham car ko push karte hai to wo easy padhta hai lekin pull karna difficult hota hai....plzzzz sir take my doubt



Yash Sharma 5 hr ago

sir ye ap physics para Raha hai kya ye advanced level ke BHI solve ho sakte hai kya plzzzz tell

"Thank you"



Harsh Raghuwanshi 1 hrs ago

sir ur teaching way is superb ,aap se padke
padai me maza aa raha hai.



AMAN AMAN 1 hrs ago

sir aap bahut accha parate ho.....LOVE YOUR
TEACHING VERY MUCH.....
PURA LECTURE ACCHE SE SAMAJH AAYA



FAB Piyash 5 hr ago

thank you sir for making physics very simpul





A particle moves from point A $(2, 1)$ to point B $(4, 4)$ along a straight line As. One of the forces acting on the particle is a variable and is given by $\vec{F} = x\hat{i} + y^2\hat{j}$. Find the work done by this force.

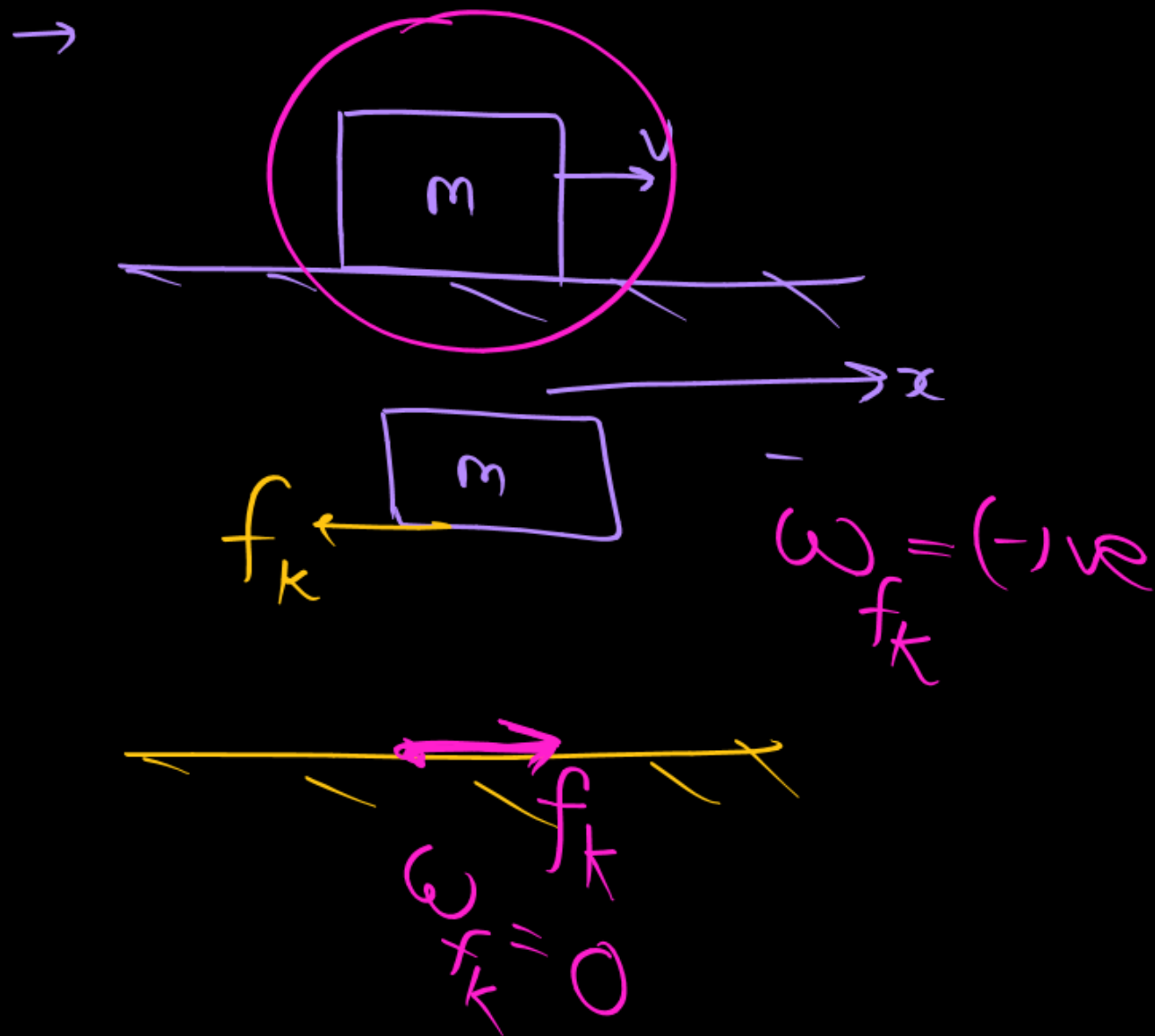
$$\begin{aligned} W &= \int_{x_1}^{x_2} F_x dx + \int_{y_1}^{y_2} F_y dy \\ &= \int_2^4 x dx + \int_1^4 y^2 dy \end{aligned}$$

$$\begin{aligned} W &= \left(\frac{x^2}{2} \right)_2^4 + \left(\frac{y^3}{3} \right)_1^4 \\ &= \left(\frac{16}{2} - \frac{4}{2} \right) + \left(\frac{4^3}{3} - \frac{1^3}{3} \right) \\ &= 6 + \frac{63}{3} = 27 \text{ J Ans} \end{aligned}$$

Thickud-Thickud

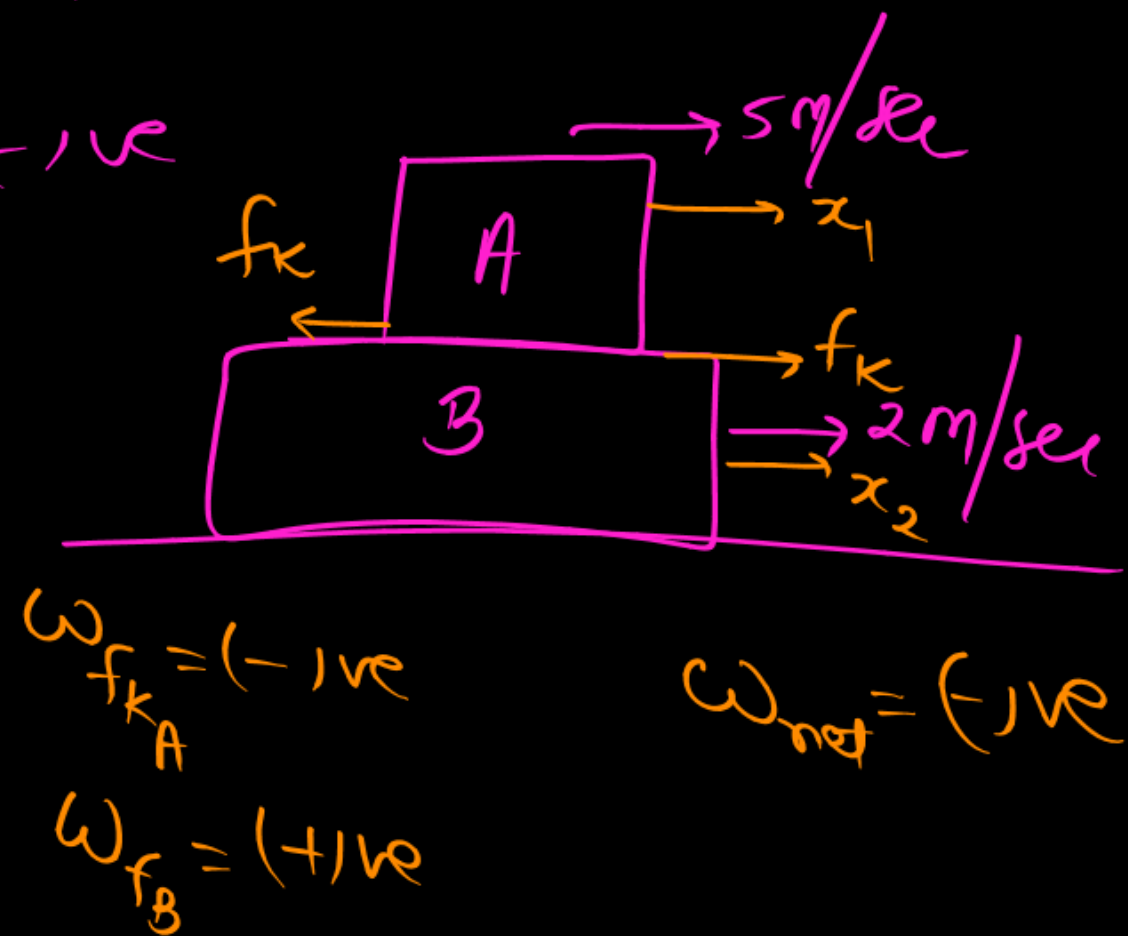
(3) Work done by friction force

(ii) Work done by kinetic friction



$$(W_{net}) = 0 + (-)ve$$

$$f_k = (-)ve$$





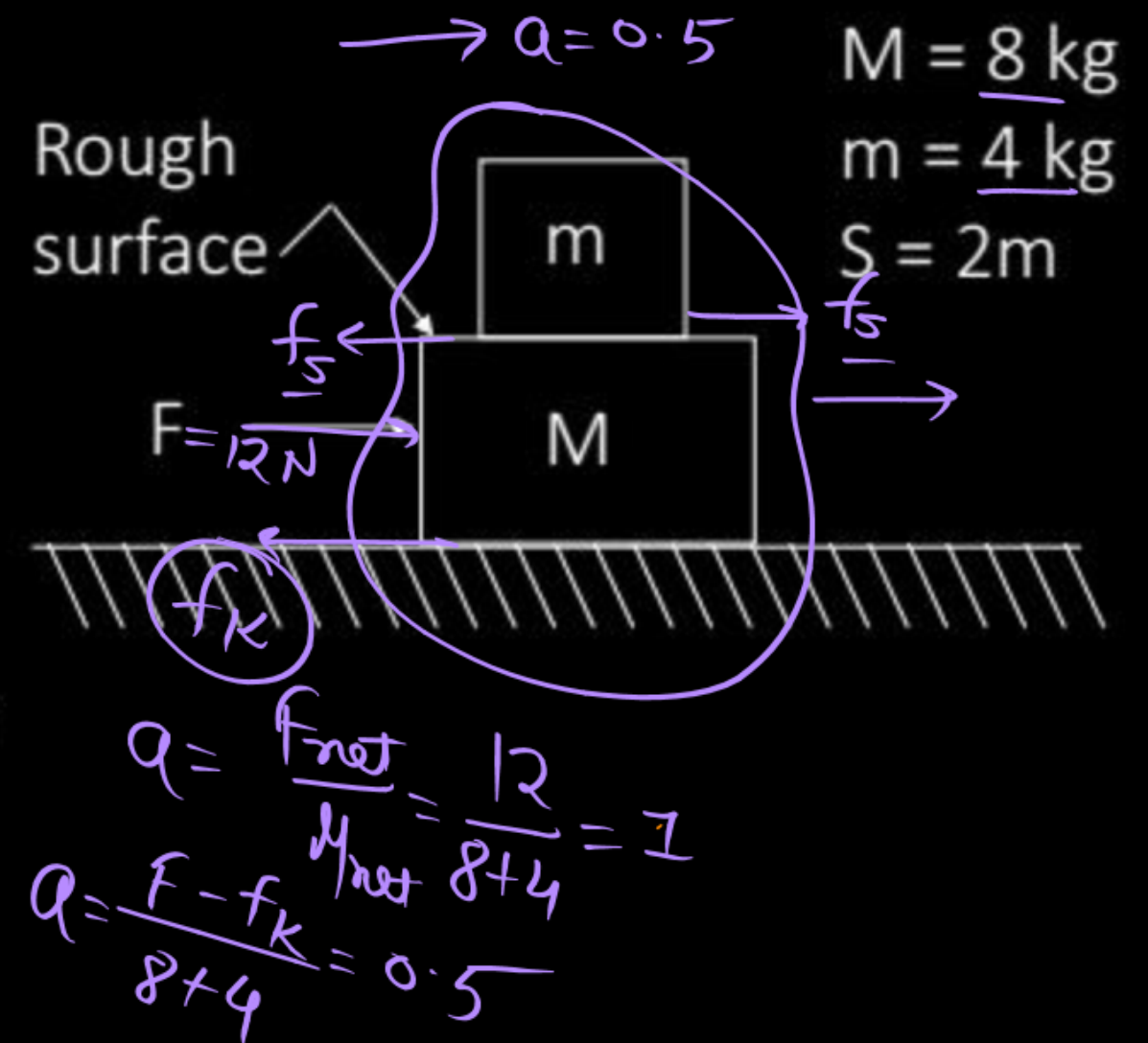
Initially the system is at rest. Now a constant horizontal force $F = 12 \text{ N}$ acting on lower block. Produces an acceleration 0.5 m/sec^2 in the system the two blocks always move together. Consider 2 meter displacement of system find out :

(i) Work done by friction acting b/w lower block. & surface on lower block.

(iii) Friction force acting on upper block.

(iv) Work done by friction force acting b/w lower block & upper block, on upper block

(v) Work done by friction force acting b/w lower block & upper block, on lower block.

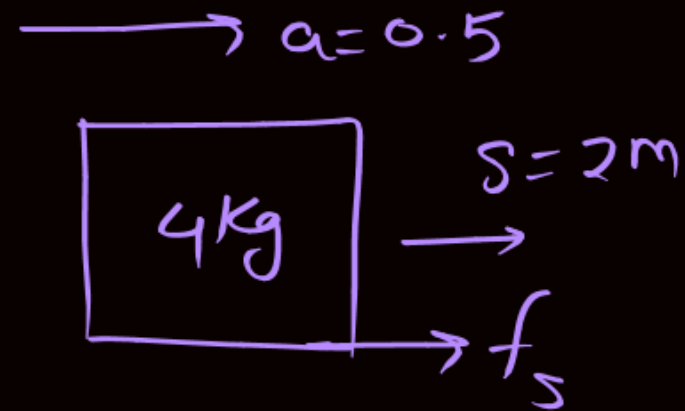


$$\frac{12 - f_k}{12} = 0.5$$

$$f_k = 6N$$

$$\begin{aligned} \rightarrow (i) \quad W &= f_k(s) \cos 180^\circ \\ &= (6)(2)(-1) \\ &= -12J \end{aligned}$$

(ii)



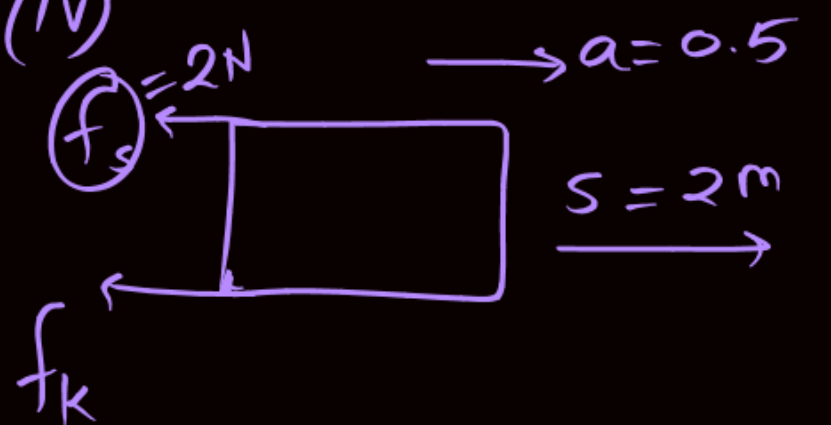
$$f_{\text{net}} = f_s = 4(0.5)$$

$$f_s = 2N$$

(iii)

$$\begin{aligned} W_{f_{s1}} &= f_s(s) \cos 0^\circ \\ &= (2)(2)(1) \\ &= 4J \end{aligned}$$

(iv)



$$\begin{aligned} W_{f_{s2}} &= (2)(2) \cos 180^\circ \\ &= -4J \end{aligned}$$

$$\begin{aligned} (W_{\text{net}})_{f_s} &= 4 - 4 \\ &= 0 \end{aligned}$$



Calculate the following parts if the system is released at $t = 0$.

H.W.

A.

Work done by the gravity on 1 kg block.

B.

Work done by gravity on 2 kg block

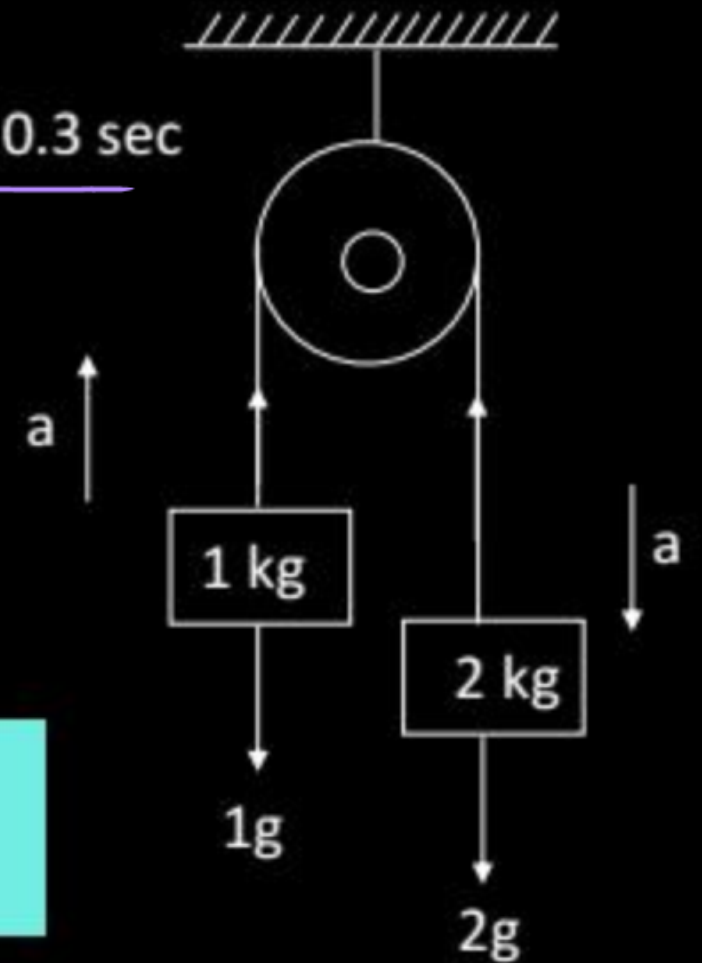
C.

Work done by tension in the string of 1 kg block

D.

Work done by tension in the string of 2 kg block

$T = 0.3 \text{ sec}$



Energy



→ Ability to do work.

→ Mechanical Energy ⇒

(i) Kinetic Energy

(ii) potential Energy

Kinetic Energy



“Kinetic energy is the form of mechanical energy possessed by a body due to its state of motion”.

$$KE = \frac{1}{2}mv^2$$

→ here $v \rightarrow$ Speed

- (i) $KE \geq 0$ always
(ii) K.E. depends on f.o.p.

(iii)

$$KE = \frac{1}{2}mv^2$$

$$\frac{d}{dt}(KE) = \frac{m}{2} \frac{d}{dt}v^2$$

$$= \frac{m}{2} \left\{ \cancel{2v} \frac{dv}{dt} \right\}$$

$$= v \{ m a_t \}$$

→ Tangential acc.

$$\frac{d}{dt}(KE) = v \{ f_t \}$$

if $KE = \text{const}$

$$\frac{d}{dt}(KE) = 0$$

$$\Rightarrow m f_t = 0$$

$$\boxed{f_t = 0}$$

Relation b/w Kinetic Energy & Momentum



$$\rightarrow KE = \frac{1}{2}mv^2$$

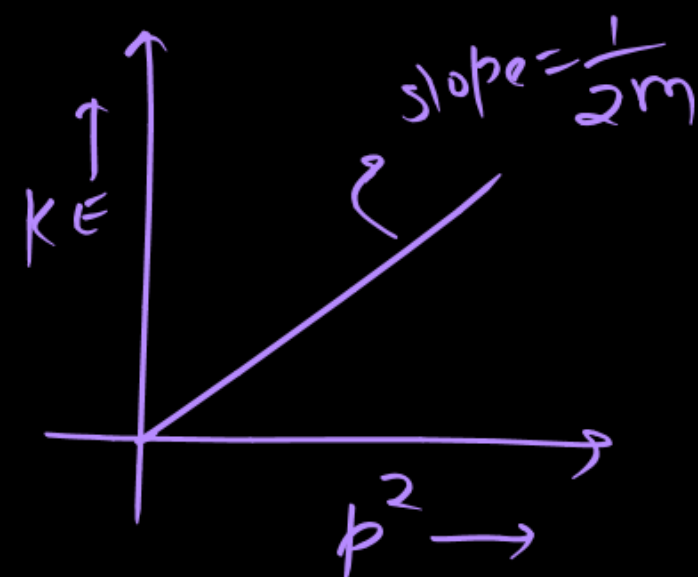
$$= \frac{(mv)^2}{2m}$$

$$\boxed{KE = \frac{p^2}{2m}}$$

$$\vec{p} = m\vec{v} \Rightarrow |\vec{p}| = p = mv$$

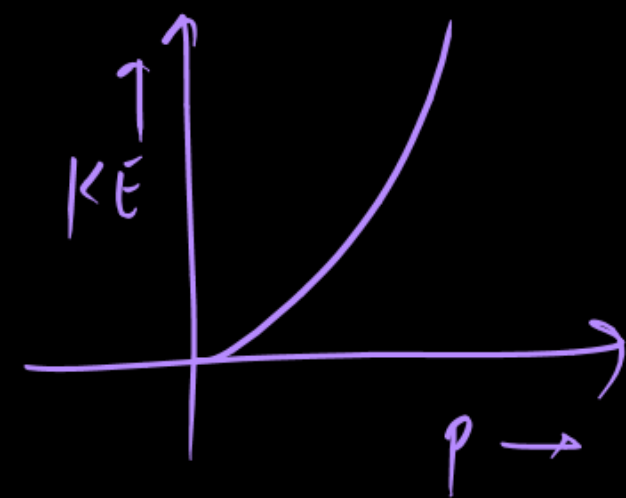
if $p \rightarrow$ pata hai $\Rightarrow$ KE. known Completely
 $m \rightarrow$ pata hai

if $KE \rightarrow$ pata hai
 $m \rightarrow$ pata hai $\Rightarrow$ p can't be known Completely



$$KE = \frac{p^2}{2m}$$

$y = (\text{slope})x$



$$KE = \frac{p^2}{2m}$$

$$\log KE = \log p^2 + \log \left(\frac{1}{2m} \right)$$

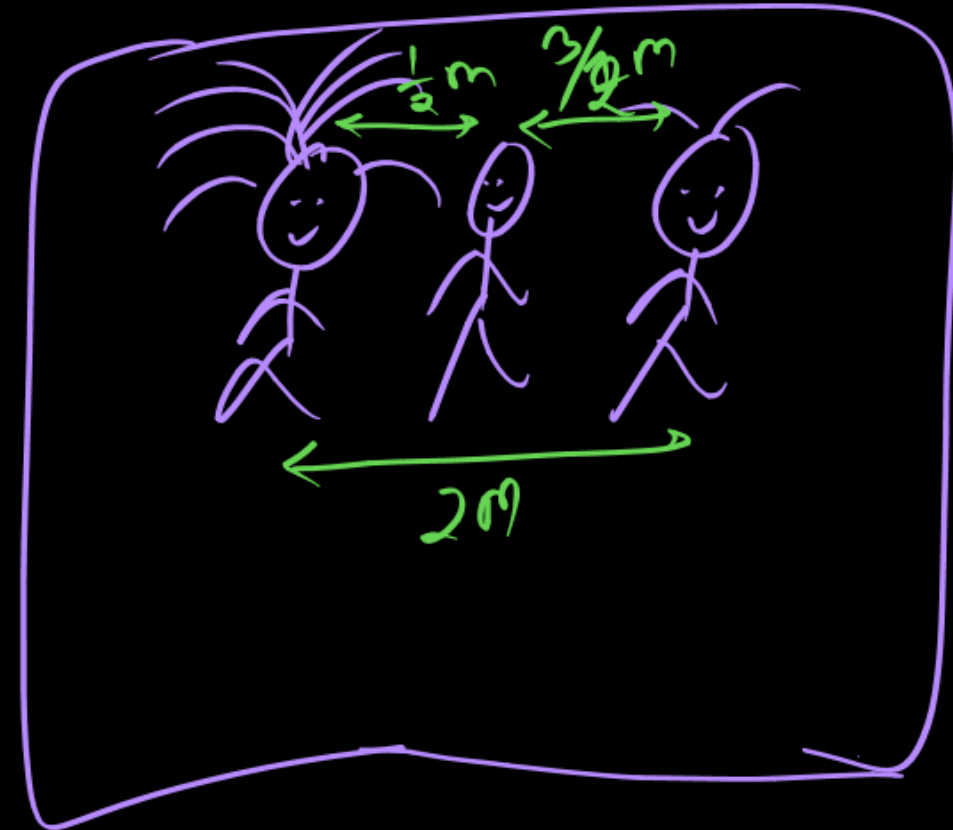
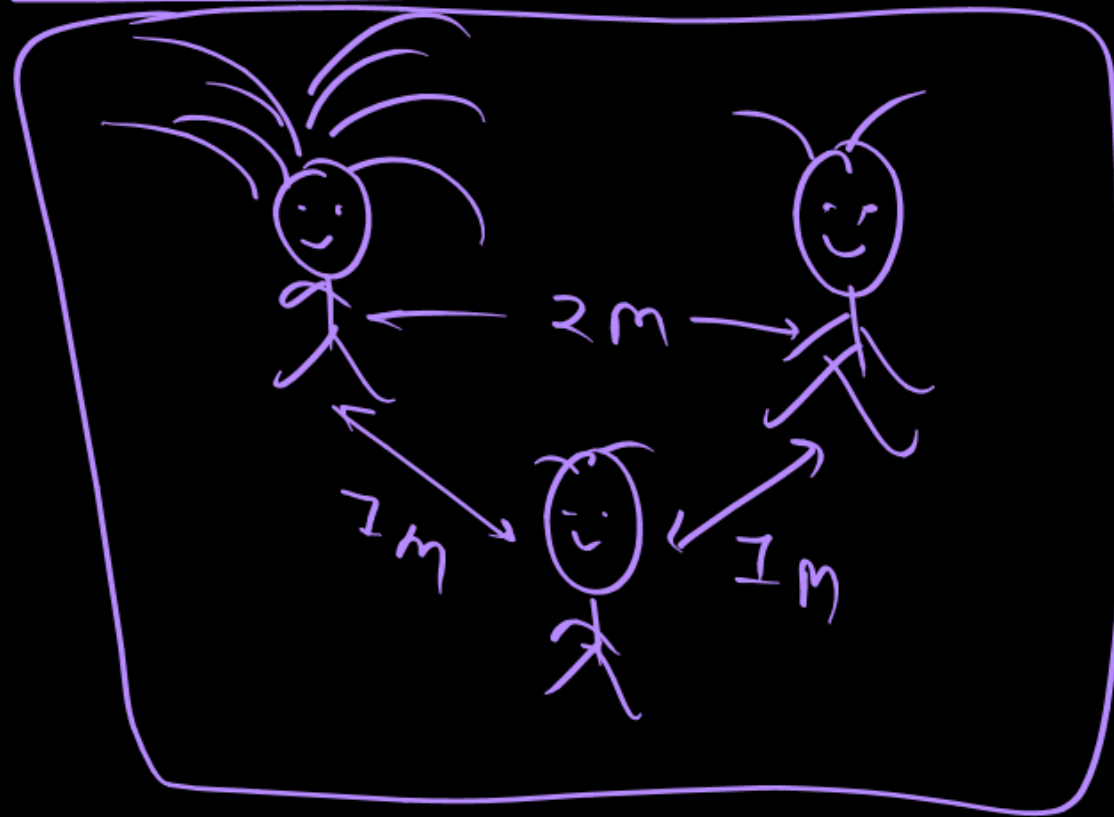
$$\log(KE) = 2 \log p - \log(2m)$$

$$y = mx - c$$

Potential energy



It is the energy which is associated with the configuration of a system of objects that exerts forces on one-another



Work – K.E. Theorem

→

$$F_{\text{net}} = m a$$

$$F_{\text{net}} = m v \frac{dv}{ds}$$

$$\int F_{\text{net}} ds = \int m v dv$$

$$\int_{s_1}^{s_2} F_{\text{net}} ds = m \int_{v_i}^{v_f} v dv$$

$$W_{\text{net}} = m \left(\frac{v^2}{2} \right)_{v_i}^{v_f}$$

$$W_{\text{net}} = m \left\{ \frac{v_f^2}{2} - \frac{v_i^2}{2} \right\}$$

$$W_{\text{net}} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$W_{\text{net}} = KE_f - KE_i$$

$$\boxed{W_{\text{net}} = \Delta KE} \quad \star \star$$

Points to remember



→ ① For different ω_{net} & ΔKE will be different but for all frames $\omega_{\text{net}} = \Delta KE$

②
$$\omega_{sp} = \left[\frac{1}{2} k x_i^2 - \frac{1}{2} k x_f^2 \right]$$

$$\omega_g = +mgh$$

$$= -mgh$$

$$= 0$$



The force $\vec{F} = 2\hat{i} + 3\hat{j}$ acting on a body having mass 5 kg and its initial position is given as $\vec{r} = \hat{i} + 2\hat{j}$ and at that time its initial speed is 10 m/s. Under the action of this force body is displaced to the final position $\vec{r}_2 = 2\hat{i} + 3\hat{j}$ then calculate the final speed of the body.

Solⁿ

$$\text{Work done} = \vec{F} \cdot \vec{S}$$

$$= (2\hat{i} + 3\hat{j}) \cdot (\vec{r}_f - \vec{r}_i)$$

$$= (2\hat{i} + 3\hat{j}) \cdot \{(2\hat{i} + 3\hat{j}) - (\hat{i} + 2\hat{j})\}$$

$$= (2\hat{i} + 3\hat{j}) \cdot (\hat{i} + \hat{j})$$

$$= (2)(1) + (3)(1) = 5 \text{ J}$$

$$W = \Delta KE$$

$$5 = \frac{1}{2} m V_f^2 - \frac{1}{2} m V_i^2$$

$$5 = \frac{1}{2} \times 5 \times V_f^2 - \frac{1}{2} \times 5 \times (10)^2$$

$$2 = V_f^2 - (10)^2$$

$$V_f^2 = 102 + 2 = 102$$

$$V_f = \sqrt{102} \text{ m/s}$$



Mass 'm' is released initially from rest. Calculate maximum extension of spring.

$$W_{\text{net}} = \Delta KE$$

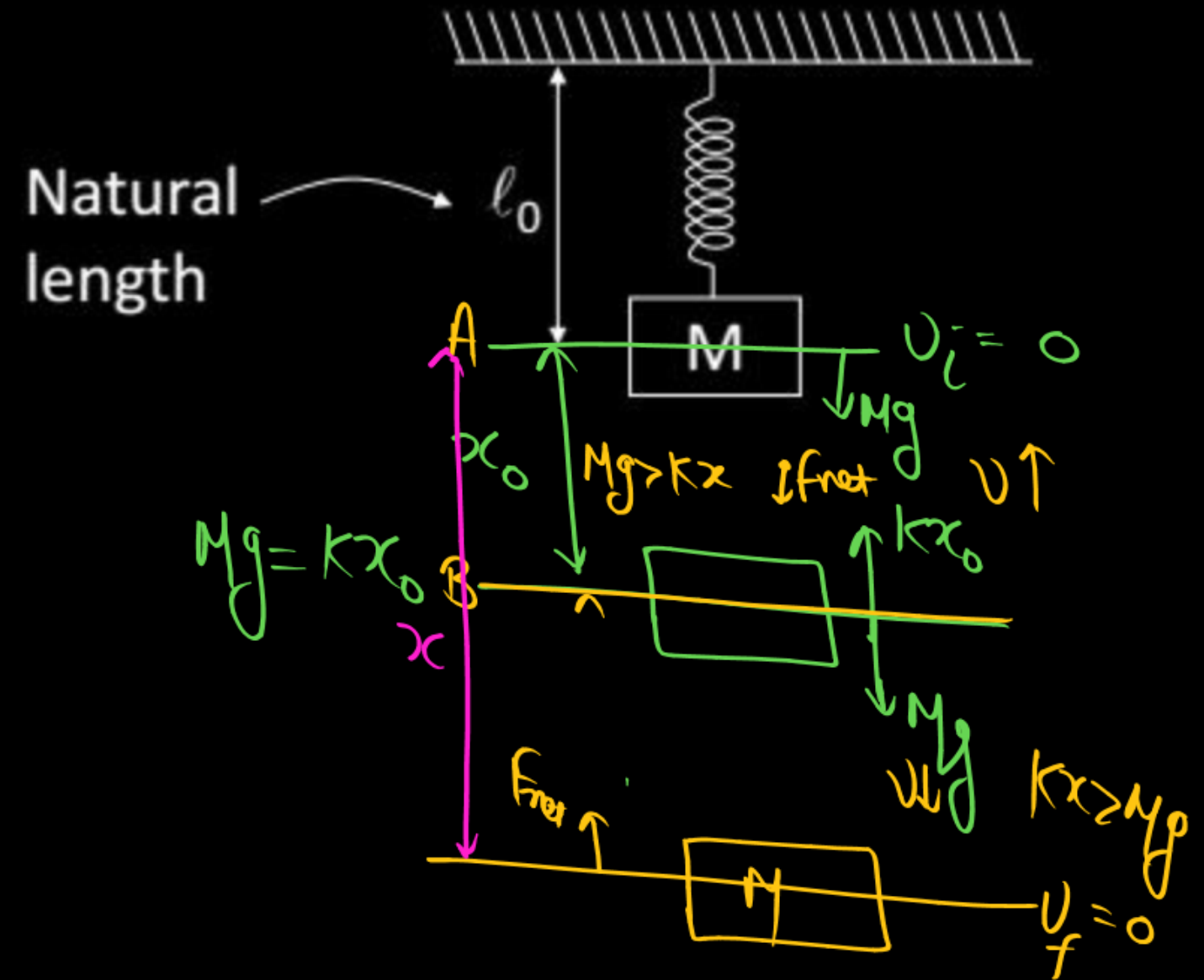
$$W_g + W_{sp} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$W_g + W_{sp} = \frac{1}{2} m (0)^2 - \frac{1}{2} m (0)^2$$

$$mgx + \left[\frac{1}{2} (kx_i^2) - \frac{1}{2} kx_f^2 \right] = 0$$

$$mgx + \frac{1}{2} k(0) - \frac{1}{2} kx^2 = 0$$

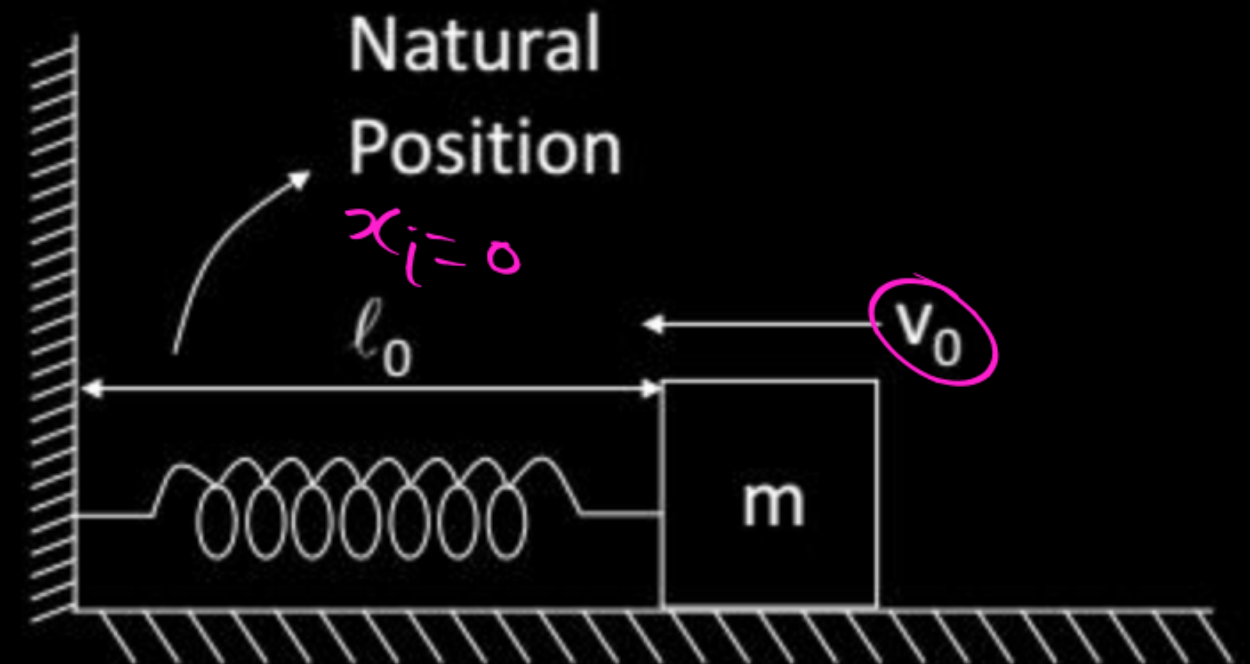
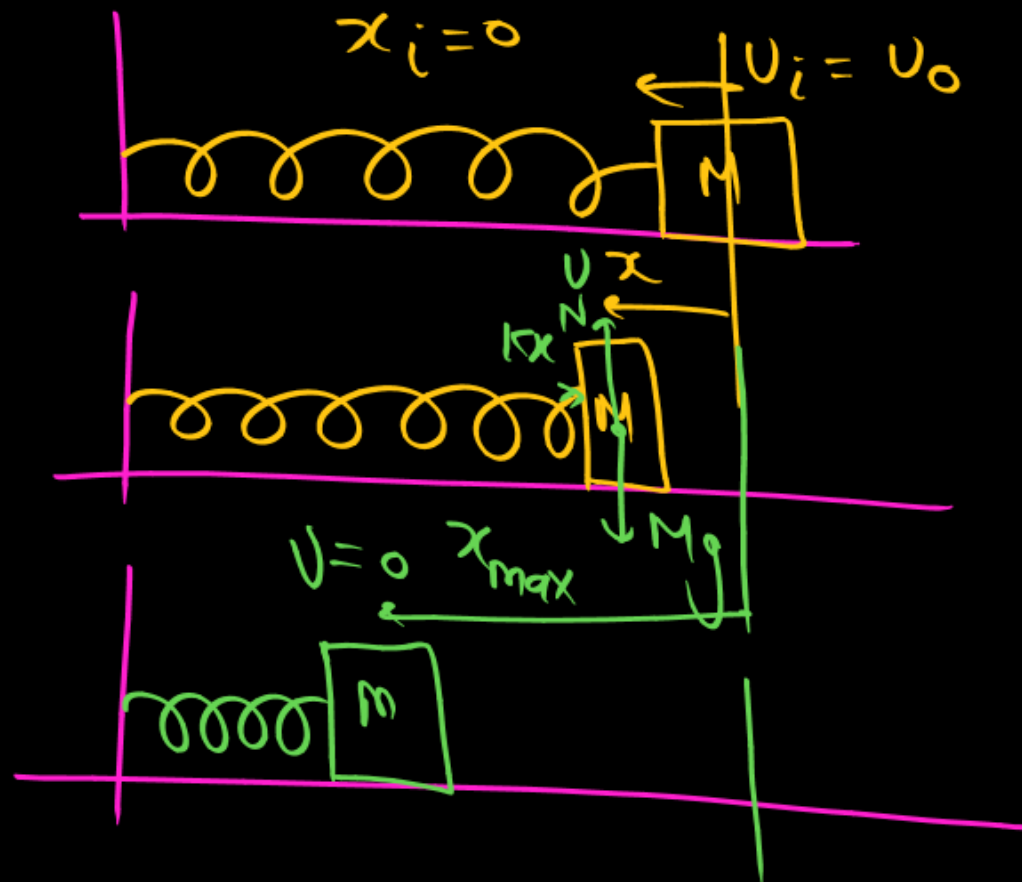
$$mgx = \frac{1}{2} kx^2 \Rightarrow \boxed{x = \frac{2mg}{k}} \quad \text{Ans.}$$





Calculate

- (i) The speed of block as the spring has been compressed by a distance 'x'.
- (ii) The maximum compression produced in the spring ?



(i)

$$W_g + W_N + W_S = \Delta KE$$

$$0 + 0 + \left\{ \frac{1}{2} k x_i^2 - \frac{1}{2} k x_f^2 \right\} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$0 + \left\{ \frac{1}{2} k (0) - \frac{1}{2} k x^2 \right\} = \frac{1}{2} m v^2 - \frac{1}{2} m v_0^2$$

$$-\frac{1}{2} k x^2 = \frac{1}{2} m (v^2 - v_0^2)$$

$$-\frac{k}{m} x^2 = v^2 - v_0^2$$

$$v^2 = v_0^2 - \frac{k}{m} x^2$$

$$v = \sqrt{v_0^2 - \frac{k}{m} x^2} \quad \text{Ans.}$$

(ii)

$$W_g + W_N + W_S = KE_f - KE_i$$

$$0 + 0 + \left\{ \frac{1}{2} k (0)^2 - \frac{1}{2} k (x_m)^2 \right\} = \frac{1}{2} m (0)^2 - \frac{1}{2} m v_0^2$$

$$-\frac{1}{2} k x_m^2 = -\frac{1}{2} m v_0^2$$

$$x_m = \left(\sqrt{\frac{3}{k}} \right) v_0 \quad \text{Ans.}$$



Calculate W.D. by a force $\vec{F} = \underbrace{ax\hat{i}}_{F_x} + \underbrace{ay\hat{j}}_{F_y}$ to move a particle from (x_1, y_1) to (x_2, y_2) .

$$\begin{aligned} W &= \int F_x dx + \int F_y dy \\ &= a \int_{x_1}^{x_2} x dx + a \int_{y_1}^{y_2} y dy \\ &= a \left(\frac{x^2}{2} \right)_{x_1}^{x_2} + a \left(\frac{y^2}{2} \right)_{y_1}^{y_2} \end{aligned}$$

$$W = a \left(\frac{x_2^2}{2} - \frac{x_1^2}{2} \right) + a \left(\frac{y_2^2}{2} - \frac{y_1^2}{2} \right) \quad \underline{\underline{Ans}}$$



Calculate W.D. by a force $\vec{F} = \underbrace{ay\hat{i}}_{F_x} + \underbrace{ax\hat{j}}_{F_y}$ to move a particle from (x_1, y_1) to (x_2, y_2) .

$$\begin{aligned}\omega &= \int F_x dx + \int F_y dy \\ &= \int y dx + \int x dy \\ &= \int (y dx + x dy) \\ &= \int d(xy)\end{aligned}$$

$$\boxed{d(uv) = u dv + v du}$$

$$\text{if } xy = z \quad z_1 = x_1 y_1 \quad z_2 = x_2 y_2$$

$$\begin{aligned}\omega &= \int dz \\ &= (z)_{z_1}^{z_2} \\ &= (xy)_{(x_1, y_1)}^{(x_2, y_2)} = (x_2 y_2 - x_1 y_1) \quad \text{Ans.} \\ &\quad \text{perfect differential form}\end{aligned}$$

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\left\{ \begin{array}{l} \frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x} \\ \frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x} \end{array} \right.$$

$$\vec{F} = \underbrace{(ay)}_{F_x} \hat{i} + \underbrace{(ax)}_{F_y} \hat{j} \quad F_z = 0$$

$$\left\{ \begin{array}{l} \frac{\partial F_x}{\partial y} = a \frac{\partial y}{\partial y} = a \\ \frac{\partial F_y}{\partial x} = a \frac{\partial x}{\partial x} = a \end{array} \right. \quad \left\{ \frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x} \right.$$

$$\left\{ \begin{array}{l} \frac{\partial F_x}{\partial z} = \frac{\partial (ay)}{\partial z} = 0 \\ \frac{\partial F_z}{\partial x} = 0 \end{array} \right. \quad \left\{ \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x} \right.$$

$$\left\{ \begin{array}{l} \frac{\partial F_y}{\partial z} = \frac{\partial (ax)}{\partial z} = 0 \\ \frac{\partial F_z}{\partial y} = 0 \end{array} \right. \quad \left\{ \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y} \right.$$

Partial differentiation $\Rightarrow$
 $p = x^2 y$

$$\begin{aligned}\frac{d}{dx}(x^2 y) &= x^2 \frac{dy}{dx} + y \frac{d(x^2)}{dx} \\ &= x^2 \frac{dy}{dx} + y(2x)\end{aligned}$$

Partial diff wrt $x \Rightarrow$

$$\begin{aligned}\frac{\partial}{\partial x}(x^2 y) &= y \frac{\partial(x^2)}{\partial x} \\ &= y(2x)\end{aligned}$$

Dependant Variable = $f(x)$
or $f(y)$
or $f(z)$

Dependant Variable = $f(x, y, z)$

Practice on partial differentiation





$$(1) \underline{[x^2 - 3x + 5]} \quad \frac{d}{dx} (x^2 - 3x + 5) = \frac{d}{dx} x^2 - 3 \frac{d}{dx} x + 0 = 2x - 3(1) + 0 = 2x - 3$$

$$(2) [x^2 yz] \rightarrow \frac{d}{dx} (x^2 yz) = yz \frac{d}{dx} x^2 = yz (2x) = 2x yz$$

$$\rightarrow \frac{d}{dy} (x^2 yz) = x^2 z \frac{d}{dy} y = x^2 z$$

$$(3) [5xz^2] \rightarrow \frac{d}{dz} (5xz^2) = 5x \frac{d}{dz} (z^2) = 5x (2z) = 10xz$$

$$(4) [y^3 z^2] \rightarrow \frac{d}{dy} (y^3 z^2) = z^2 \frac{d}{dy} y^3 = z^2 (3y^2) = 3y^2 z^2$$



H.W.



Calculate W.D. by a force F to move a particle from $(0, 0)$ to $(2, 4)$ here $\vec{F} = 2xy^2\hat{i} + 2yx^2\hat{j}$



If force $\vec{F} = ay\hat{i}$

Then calculate W.D. by force $\vec{F}$ to displace a particle from (x_1, y_1) to (x_2, y_2) .

$$\vec{F} = ay\hat{i}$$

$$W = \int_{x_1}^{x_2} F_x dx = a \int_{x_1}^{x_2} y dx$$

if we are given Relation b/w y & x

Matlab $y = f(x)$

$$\begin{cases} \frac{\partial F_x}{\partial y} = \frac{d}{dy}(ay) = a \\ \frac{\partial F_y}{\partial x} = 0 \end{cases}$$

$$\frac{\partial F_x}{\partial y} \neq \frac{\partial F_y}{\partial x}$$

Relation b/w y & x

$\Rightarrow$ Curve eqⁿ

Path of particle

$y = x^2$
 $y = 2x + 3$

Conservative force



$$W = \int_{s_1=s}^{s_2=s} F(ds)$$

$$(\text{Upper limit} - \text{Lower limit})$$

- (1) W.D. by conservative force depends only on initial & final position and not on the path.
- (2) W.D. by Conservative force for a closed loop is zero.
- (3) Whenever these forces acts, they never dissipates energy i.e. total mechanical energy (KE + PE) of system remains conserved.

Non- conservative force

- ✓ (1) W.D. by Non-conservative forces depends on path followed by body.
- ✓ (2) W.D. by Non-Conservative forces in a closed loop is Non-zero.
- ✓ (3) A non-conservative force acting with in a system causes a change in mechanical energy (K.E. + P.E.) of the system when $W_{\text{non cons}} \neq 0$.

friction



A force acting on the particle is given as $\vec{F} = x^2\hat{i} + y^2\hat{j}$. Then calculate the work done if the particle is taken across the path which is given as shown.

$$W = 0$$

Sol^y

$$\vec{F} = \underbrace{x^2}_{f_x}\hat{i} + \underbrace{y^2}_{f_y}\hat{j} \quad f_z = 0$$

$$\frac{\partial f_x}{\partial y} = \frac{\partial f_y}{\partial x}$$

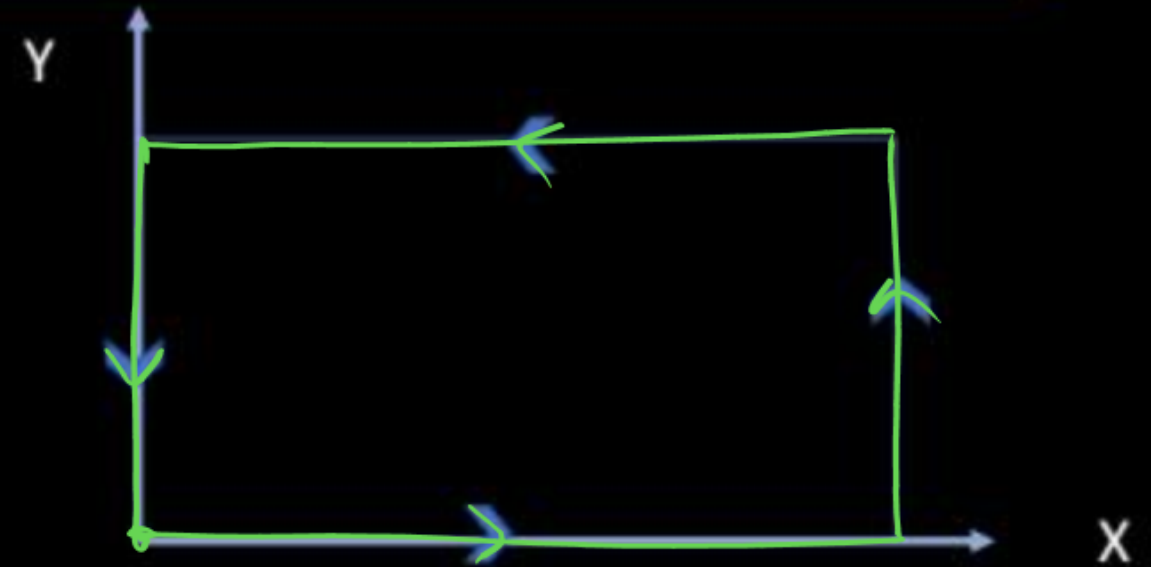
$$\frac{\partial}{\partial y}(x^2) = \frac{\partial}{\partial x}(y^2)$$

$$x^2 \frac{\partial(1)}{\partial y} = y^2 \frac{\partial(1)}{\partial x}$$

$$0 = 0$$

$$\frac{\partial f_x}{\partial z} = \frac{\partial f_z}{\partial x}$$
$$0 = 0$$

$$\frac{\partial f_y}{\partial z} = \frac{\partial f_z}{\partial y}$$
$$0 = 0$$





H.W.



The force acting on the particle is given as $\vec{F} = \hat{i} + y\hat{j}$ and the particle is displaced from point O to the final point B through two given path as shown in the figure

A.

$$W_I > W_{II}$$

B.

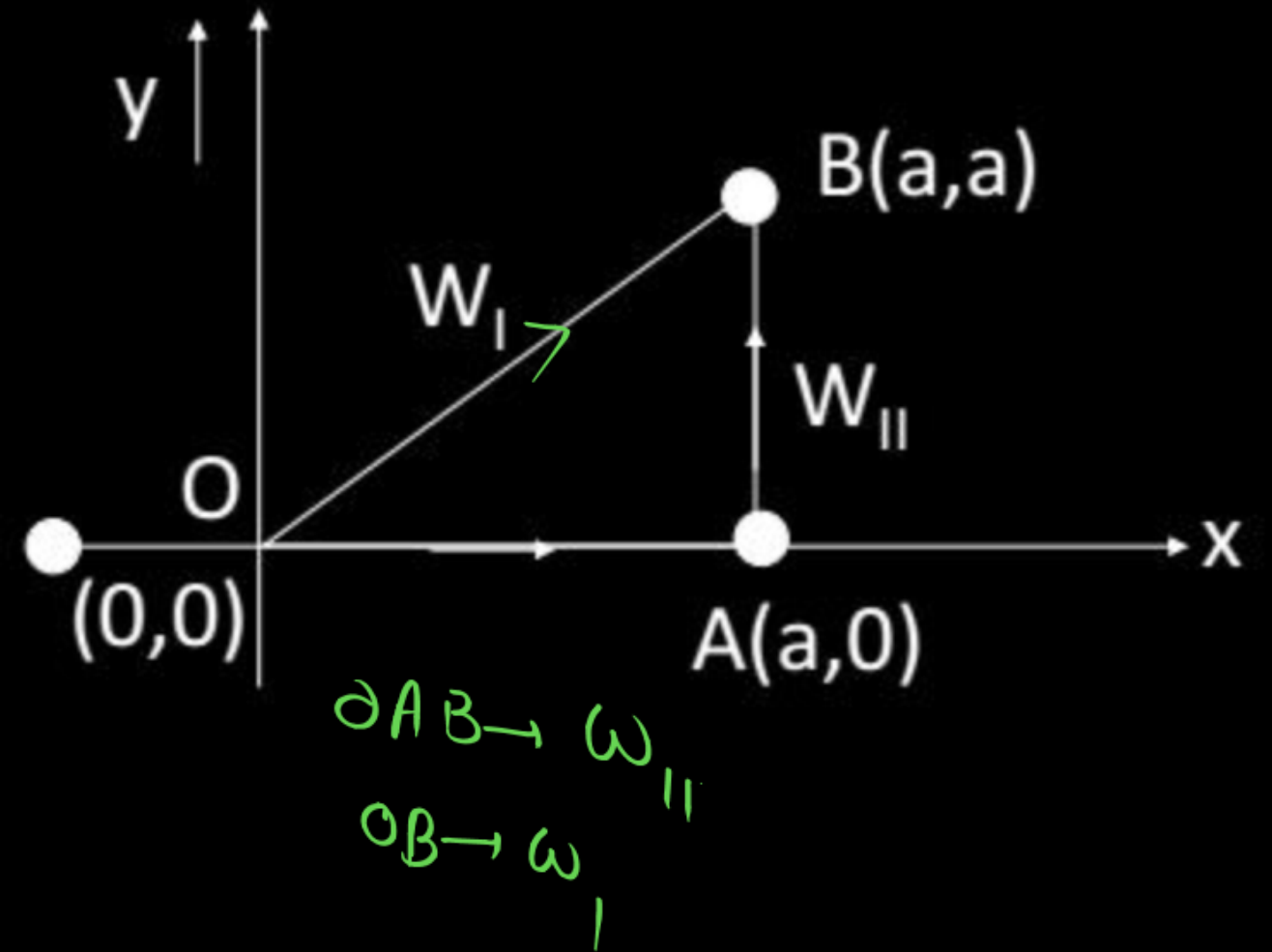
$$W_{II} > W_I$$

C.

$$W_I = W_{II}$$

D.

Do not Know



LAGE RAHO.....

ME TUJHE EK DIN JARROR MILUNGI

(AAPKI APNI IIT SEAT)

THANK - YOU

